1. (a) Trobem directament

$$P_{\text{útil}} = P_{\text{elèctrica}} \cdot \eta_m \eta_r = 4,5 \cdot 0,88 \cdot 0,92 = 3{,}643\,\text{kW}$$

(b) De forma semblant a l'apartat anterior

$$P_2 = P_{\text{útil}} \cdot \eta_1 = 3,96 \cdot 0,92 = 3{,}643\,\text{kW}$$

Fem un factor de conversió per la velocitat angular

$$120\,\frac{\text{rev}}{\text{min}} \cdot \frac{2\pi\,\text{rad}}{1\,\text{rev}} \cdot \frac{1\,\text{min}}{60\,\text{s}} = 4\pi\,\text{rad/s}$$

i el parell es pot calcular com

$$\Gamma_2 = \frac{P_2}{\omega_2} = \frac{3643}{4\pi} = 289{,}9\,\text{Nm}$$

(c) Trobem directament

$$\tau = \frac{\omega_2}{\omega_1} = \frac{n_2}{n_1} = \frac{120}{1440} = 0,083$$

2. (a) En primer lloc calculem

$$P_1 = \Gamma_1 \omega_1 = 20 \cdot \frac{2\pi}{60} = 1000\pi = 3141{,}6\,\text{W} = 3{,}142\,\text{kW}$$

on hem fet implícitament el factor de conversió per la velocitat angular. ara calculem

$$P_2 = P_1 \eta = 3141,6 \cdot 0,95 = 2984{,}5\,\text{W} = 2{,}985\,\text{kW}$$

(b) Podem escriure

$$\omega_1 d_1 = \omega 2 d_2 \rightarrow n_1 d_1 = n_2 d_2 \rightarrow n_2 = n_1 \frac{d_1}{d_2} = 1500 \cdot \frac{100}{250} = 600\,\text{min}^{-1}$$

(c) El càlcul del parell és immediat

$$\Gamma_2 = \frac{P_2}{\omega_2} = \frac{2984,5}{20\pi} = 47{,}5\,\text{Nm}$$

on hem tingut en compte que

$$600\,\frac{\text{rev}}{\text{min}} \cdot \frac{2\pi\,\text{rad}}{1\,\text{rev}} \cdot \frac{1\,\text{min}}{60\,\text{s}} = 20\pi\,\text{rad/s}$$

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3. (a) Amb un factor de conversió

$$28\,\frac{\text{MJ}}{kg}\cdot\frac{45\,\text{t}}{\text{h}}\cdot\frac{1000\,\text{kg}}{1\,\text{t}}\cdot\frac{1\,\text{h}}{3600\,\text{s}}=3{,}5\cdot10^5\,\text{MW}=350\,\text{MW}$$

(b) El rendiment demanat es pot calcular com

$$\eta=\frac{110}{350}=0,314$$

(c) Podem escriure

$$110\,\text{MW}=110\cdot10^3\,\text{kW}$$

i ara

$$110\cdot10^3\,\text{kW}\cdot24\,\text{h}=2{,}64\cdot10^3\,\text{kWh}$$

L'energia demanada serà

$$110\cdot10^6\cdot24\cdot3600\,\text{J}\cdot\frac{1\,\text{MJ}}{1\cdot10^6\,\text{J}}\cdot\frac{1\,\text{kg}}{28\,\text{MJ}}=339\,428{,}57\,\text{kg}$$

en realitat, degut al rendiment, la quantitat serà més gran

$$\frac{339428,57}{\eta}=1{,}081\cdot10^6\,\text{kg}=1{,}081\cdot10^3\,\text{t}$$

4. (a)

$$\frac{220\,\text{MW}}{\eta}=\frac{220\cdot10^6}{0,55}=400\cdot10^6=4\cdot10^8\,\text{W}$$

en 1 s

$$4\cdot10^8\,\frac{\text{J}s}{\cdot}\,\frac{1\,\text{kg}}{48\cdot10^6\,\text{J}}=8,33\,\frac{\text{kg}}{s}$$

(b) Mitjançant un factor de conversió

$$8,33\,\frac{\text{kg}}{\text{s}}\cdot\frac{3600\,\text{s}}{1\,\text{s}}\cdot\frac{1\,\text{m}^3}{0{,}8\,\text{kg}}=37\,485\,\text{m}^3/\text{h}$$

(c) Finalment

$$\frac{1\,\text{kWh}}{\eta}=\frac{1\,\text{kWh}}{0,55}=1{,}818\,\text{kWh}$$

llavors

$$1{,}818\,\text{kWh}\cdot\frac{0,045\,€}{1\,\text{kWh}}=0,082\,€$$

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